

A $m > n$: split the columns, W_2 is $n \times n$

k_proj / v_proj of Qwen2.5-1.5B: $n = 256$, $m = 1536$

block split

factorization

$$\begin{matrix} n \\ \text{---} \end{matrix} \begin{matrix} \text{---} W_1 \text{---} W_2 \text{---} \\ m-n \quad n \end{matrix} \cdot \begin{matrix} x_1 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_m \end{matrix} = \begin{matrix} y_1 \\ \vdots \\ y_n \\ y_{n+1} \\ \vdots \\ y_m \end{matrix} \quad W_0 = \begin{matrix} W_2 \end{matrix} \cdot \begin{matrix} \text{---} \overline{W} \text{---} \\ \vdots \end{matrix}$$

$$\text{solve } W_2 \overline{W} = W_1 \Rightarrow \overline{W} = W_2^{-1} W_1$$

$$W_0 x = W_1 x_1 + W_2 x_2 = W_2 (\overline{W} x_1 + x_2)$$

$$\text{freeze } \overline{W}, \text{ train } W_2: \Delta W = \Delta W_2 [\overline{W} \mid I] \Rightarrow \text{row}(\Delta W) \subseteq \text{row}(W_0)$$

B $m < n$: split the rows, W_2 is $m \times m$

gate_proj / up_proj of Qwen2.5-1.5B: $n = 8960$, $m = 1536$

block split

factorization

$$\begin{matrix} n-m \\ \text{---} \end{matrix} \begin{matrix} W_1 \\ \vdots \\ W_2 \end{matrix} \cdot \begin{matrix} x \\ \vdots \\ y_2 \end{matrix} = \begin{matrix} y_1 \\ \vdots \\ y_m \end{matrix} \quad W_0 = \begin{matrix} \overline{W} \\ \vdots \\ I \end{matrix} \cdot \begin{matrix} W_2 \end{matrix}$$

$$\text{solve } \overline{W} W_2 = W_1$$

$$\Rightarrow \overline{W} = W_1 W_2^{-1}$$

$$W_0 x = [W_1 x; W_2 x]$$

$$= [\overline{W}; I] W_2 x$$

$$\text{freeze } \overline{W}, \text{ train } W_2: \Delta W = [\overline{W}; I] \Delta W_2 \Rightarrow \text{col}(\Delta W) \subseteq \text{col}(W_0)$$

 W_2 : trainable

 $W_1, \overline{W}$: frozen

 I : index selection